

The Lunar Mass Driver

Throwing Cargo off the Moon with Electricity Instead of Rockets

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Abstract

A rocket carries its own violence: to leave a world it must burn propellant it brought with it, and most of what a rocket lifts is the fuel to lift the fuel. On a small, airless world like the Moon, there is a better way to send cargo into space — one that leaves the energy source on the ground and the propellant out of the equation entirely. This study describes the Lunar Mass Driver, an electromagnetic catapult that accelerates payloads along a track using magnetic forces and flings them off the Moon at the speed needed to reach orbit or beyond. Because the Moon's gravity is weak and it has no atmosphere to fight, the required speed is modest and nothing is lost to air; because the accelerator is fixed to the ground and powered by electricity, no propellant rides with the cargo. The idea is not new — it was studied in detail decades ago as the key to exporting lunar material cheaply — and it rests on the same magnetic-launch physics now maturing on Earth. We describe the principle, the machine, what it is and is not good for, and the honest engineering of launching solid cargo by electromagnetism.

1. Introduction: The Tyranny of the Rocket, and an Escape from It

Every rocket that has ever flown has been shackled by the same arithmetic. To gain speed it throws propellant backward, and to carry that propellant it needs more propellant, so that the fuel required grows steeply with the speed sought. The result is that a rocket leaving Earth is almost entirely fuel, with only a sliver of payload atop a towering tank. This is the tyranny of the rocket equation, and it is why spaceflight from Earth is so costly. On Earth we have little choice, because Earth's deep gravity and thick atmosphere leave few alternatives for reaching orbit.

The Moon is different in exactly the ways that matter. Its gravity is about a sixth of Earth's, so the speed needed to leave its surface for orbit is far lower; and it has no atmosphere, so there is no air resistance and no need for streamlining or for climbing above the thickest air. On such a world a wholly different launch method becomes practical: instead of carrying energy as fuel, leave the energy on the ground as electricity, and use it to accelerate the cargo along a fixed track by magnetic force, releasing it at the modest speed the Moon's weak gravity requires. The rocket's tyranny is broken not by a better rocket but by dispensing with the rocket altogether. The machine that does this is the mass driver.

2. Concept Description

The Lunar Mass Driver is a long, straight electromagnetic accelerator — in effect a catapult that uses magnetism instead of tension. A payload is placed in a carrier, or bucket, that runs along a track lined with electromagnets. As the bucket passes, the magnets are switched in sequence to pull it ever faster, building speed over the length of the track until the payload is moving fast enough to coast away from the Moon. At the end of the track the payload is released to fly free, while the bucket is braked and returned to the

start to be loaded again. No part of the launch energy leaves with the cargo; it is all delivered electrically, from the track, during the run.

Because the Moon's escape and orbital speeds are low and there is no air, the track need not be impossibly long or the accelerations impossibly violent, though both are substantial. The mass driver is best understood as an exporter of bulk material and rugged cargo: raw regolith, processed metals, blocks of material, propellant, or anything that can withstand a hard, brief acceleration and does not mind being flung. It is not a launcher for delicate cargo or for people, who could not survive the accelerations a compact track imposes. Its purpose is to make the launch of tough lunar material cheap and repeatable, turning the Moon into a supplier of mass to the rest of cislunar space (O'Neill, 1974, on the mass driver as the basis for using lunar material).

3. The Physics of an Electromagnetic Launch

The principle is electromagnetic acceleration: a series of electromagnets, energised in a travelling sequence, exerts force on a moving carrier and speeds it along a track — the same principle as the linear motors that drive maglev trains and the electromagnetic catapults now launching aircraft, scaled up to launch speed. Over a long enough track, even a comfortable force sustained for the whole length can build very high speed, because speed accumulates with distance; the longer the track, the gentler the acceleration needed to reach a given launch speed, and the trade between track length and acceleration is the central design choice.

On the Moon the numbers are forgiving. The speed needed to reach lunar orbit or to escape the Moon is a small fraction of what it takes to leave Earth, because the Moon's gravity is weak; and with no atmosphere, the payload loses nothing to drag and needs no protection from heating, so the launch speed at the end of the track is very nearly the speed the payload keeps. Electricity to run the accelerator can come from solar arrays or other surface power, so the energy source is abundant and stationary. The physics is entirely conventional — magnetic force, kinematics, orbital mechanics — and contains no unknowns; the mass driver asks only for engineering at scale, not for any new science (O'Neill, 1974).

4. The Machine and Its Limits

A mass driver comprises a long track carrying a sequence of drive coils or magnets, precisely timed switching to energise them as the bucket passes, a reusable bucket that grips the payload during acceleration and releases it cleanly at the end, a braking and return system to recover the bucket, and a substantial electrical supply with the means to deliver power in the intense, brief pulses each launch demands. Aiming matters: because the payload coasts on a ballistic trajectory after release, the launch direction and speed must be controlled precisely so that the cargo arrives where intended, whether a collection point in orbit or a particular destination.

The limits define the tool. The accelerations required to launch from a track of practical length are far too high for people or fragile equipment, so the mass driver is strictly a freight launcher for durable cargo. Its output is a stream of ballistic packages that must be caught, gathered, or guided at the far end, implying a receiving system in orbit or at the destination. And it launches only from the Moon outward; it does nothing for landing, which needs other means. Within those limits it is extraordinarily powerful: once built and powered, it can throw cargo off the Moon again and again at the cost only of electricity, with no propellant consumed and nothing expended but the wear of the machine — a launch economy impossible

for any rocket.

Aspect	Rocket launch	Mass driver
Energy carried	as propellant, onboard	as electricity, on ground
Payload fraction	small	high (cargo only)
Per-launch cost	expended fuel/stage	electricity + wear
Good for	any cargo, people	durable bulk cargo only
Needs	fuel supply chain	power + a catcher in orbit

Table 1. The mass driver against the rocket for lunar export.

5. Operations Concept

Phase one is a short demonstrator: a modest track proving the accelerate-release-catch cycle at low launch mass, validating the coil switching, the bucket recovery, and the aiming, without yet reaching full export scale. Phase two is a production launcher exporting bulk lunar material — regolith, processed metal, or propellant feedstock — to a receiving station in lunar orbit, turning the Moon into a steady supplier of mass to cislunar space. Phase three integrates the mass driver into a wider lunar industrial economy in which material mined and processed on the surface is launched continuously and cheaply to where it is needed, so that the cost of moving mass off the Moon ceases to be a limiting factor. Each phase depends on a receiving system able to gather the launched packages, making the mass driver and its orbital catcher a single coupled system.

6. Honest Difficulties

The challenges are of engineering and scale, not physics. Delivering the large, precisely-timed pulses of electrical power that each launch requires is demanding, as is supplying the average power for a high launch rate. The release must be clean and the aim exact, because a payload coasting ballistically cannot correct itself and small errors at release become large errors downrange, so the launcher must be finely controlled and a receiving system must gather the arriving cargo. The accelerations rule out people and fragile goods entirely, restricting the mass driver to rugged freight. Building and aligning a long, precise track on the lunar surface, and maintaining it against dust and thermal extremes, is a serious construction task. And the whole scheme is useful only where there is something worth launching in quantity and something to catch it — it is infrastructure for an established lunar industry, not a first step. None of this is a barrier of physics; the launch principle is the same one maturing in terrestrial electromagnetic launchers, and the lunar application was worked out in detail decades ago (O'Neill, 1974). They are the real costs of building a planet-scale catapult.

7. Pathway to Reality

The concept is unusually well-founded for so dramatic a machine. Electromagnetic launch is real and advancing on Earth, from maglev to aircraft catapults, so the core technology of accelerating a carrier magnetically along a track is established and improving; and the specific application of a mass driver to launching lunar material was studied quantitatively in the 1970s as the linchpin of using the Moon's resources (O'Neill, 1974). The credible path runs from a terrestrial or short lunar demonstrator through to a production launcher, and it becomes compelling exactly when there is lunar material worth exporting in

bulk — which is to say, when the mining, refining, and manufacturing concepts elsewhere in this catalogue have given the Moon something to sell. The mass driver is the Moon's loading dock, and it becomes worth building the moment the Moon becomes a producer.

8. Conclusion

The rocket's great weakness is that it must lift the means of its own ascent, and on a small airless world that weakness can simply be set aside. The Lunar Mass Driver leaves the energy on the ground, dispenses with propellant altogether, and throws cargo off the Moon with nothing but electricity and magnetism, again and again, for the price of the power it draws. It cannot launch people and it cannot land anything, and it is only worth building once the Moon has goods to ship; but for the steady export of rugged lunar material — the metal, the regolith, the propellant that a spacefaring economy will want in quantity — it offers a launch cost that no rocket can approach. It is the moment the Moon stops being only a destination and becomes a source, flinging its wealth outward on a beam of electric force.

References

Harvard referencing style (Cite Them Right).

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